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AOOP-Week-1

1. **Create a class Student with student ID, name and age. Create three student objects and display their details.**

Student.java > Student

```java
class Student {
    int studentId = 101;
    String name = "Srithika";
    int age = 19;

    void display() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }

    Run | Debug
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.display();
    }
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SPELL CHECKER 1  AZURE  Run: Student

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C:\Users\knsat\OneDrive\SRITHIKA\Java essages -cp "C:\Users\knsat\AppData\Roaming\Code\User\workspaceStorage\8da20085
f1bd8fde420b8550acc4cb32\redhat.java\jdt_ws\Java practice_284cc92b\bin" Student "
Student ID: 101
Name: Srithika
Age: 19

C:\Users\knsat\OneDrive\SRITHIKA\Java practice>
```

2. **Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.**

Employee.java > Employee > main(String[])

```java
class Employee {
    int employeeId;
    String name;
    double salary;

    // Parameterized constructor
    Employee(int id, String n, double s) {
        employeeId = id;
        name = n;
        salary = s;
    }

    // Method to display employee details
    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
        System.out.println();
    }

    Run | Debug
    public static void main(String[] args) {
        Employee e1 = new Employee(id: 101, n: "Rahul", s: 30000);
        Employee e2 = new Employee(id: 102, n: "Srithika", s: 35000);

        e1.display();
        e2.display();
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 3 AZURE

Run: Student
Run: Emplo...

```
able-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\knsat\AppData\Roaming\Code\User\workspaceStorage\8
da20085f1bd8fde420b8550acc4cb32\redhat.java\jdt_ws\Java practice_284cc92b\bin" Employee "
Employee ID: 101
Name: Rahul
Salary: 30000.0

Employee ID: 102
Name: Srithika
Salary: 35000.0
```

3. **Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.**

BankAccount.java

```java
class BankAccount {
    int accountNumber;
    double balance;

    // Constructor
    BankAccount(int accNo, double bal) {
        accountNumber = accNo;
        balance = bal;
    }

    // Deposit method
    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }

    // Withdrawal method
    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println(x: "Insufficient balance");
        }
    }

    // Display account details
    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Current Balance: " + balance);
    }

    Run | Debug
    public static void main(String[] args) {
        BankAccount b1 = new BankAccount(accNo: 12345, bal: 5000);

        b1.deposit(amount: 2000);
        b1.withdraw(amount: 1000);

        b1.display();
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 3 AZURE

```
C:\Users\knsat\OneDrive\SRITHIKA\Java practice> c: && cd "c:\Users\knsat\OneDrive\SRITHIKA\Java practice" &&
cmd /C "C:\Users\knsat\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\latest\b
in\java.exe --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\knsat\AppData\Roaming\Code
\User\workspaceStorage\8da20085f1bd8fde420b8550acc4cb32\redhat.java\jdt_ws\Java practice_284cc92b\bin" BankAc
count "
Deposited: 2000.0
Withdrawn: 1000.0
Account Number: 12345
Current Balance: 6000.0
```

4. **Create a Rectangle class with length and width. Calculate and display its area and perimeter.**

Rectangle.java > Rectangle

```java
class Rectangle {
    double length;
    double width;

    // Constructor
    Rectangle(double l, double w) {
        length = l;
        width = w;
    }

    // Calculate area and perimeter
    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Length: " + length);
        System.out.println("Width: " + width);
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    Run | Debug
    public static void main(String[] args) {
        Rectangle r1 = new Rectangle(l: 10, w: 5);
        r1.calculate();
    }
}
```

```
C:\Users\knsat\OneDrive\SRITHIKA\Java practice>
C:\Users\knsat\OneDrive\SRITHIKA\Java practice> c: && cd "c:\Users\knsat\OneDrive\SRITHIKA\Java practice" && cmd /C "
C:\Users\knsat\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\latest\bin\java.exe --en
able-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\knsat\AppData\Roaming\Code\User\workspaceStorage\8
da20085f1bd8fde420b8550acc4cb32\redhat.java\jdt_ws\Java practice_284cc92b\bin" Rectangle "
Length: 10.0
Width: 5.0
Area: 50.0
Perimeter: 30.0
```

5. **Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.**

Book.java > Book

```java
class Book {
    String ISBN;
    String title;
    double price;

    // Parameterized constructor
    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    // Method to display book details
    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    Run | Debug
    public static void main(String[] args) {
        Book b1 = new Book(i: "9781234567890", t: "Java Programming", p: 450);

        b1.display();
    }
}
```

```
C:\Users\knsat\OneDrive\SRITHIKA\Java practice>
C:\Users\knsat\OneDrive\SRITHIKA\Java practice> c: && cd "c:\Users\knsat\OneDrive\SRITHIKA\Java practice" && cmd /C "
C:\Users\knsat\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\latest\bin\java.exe --en
able-preview -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\knsat\AppData\Roaming\Code\User\workspaceStorage\8
da20085f1bd8fde420b8550acc4cb32\redhat.java\jdt_ws\Java practice_284cc92b\bin" Book "
ISBN: 9781234567890
Title: Java Programming
Price: 450.0
```